



Gandaki University
Faculty of Science and Technology
Bachelor of Information Technology (BIT)

Project Progress Interaction Log Sheet

Academic Year: 2022 Semester: 8th Visit No: 3 Date of Visit:

Project Title: Dhoom Car Racing

Student Name(s): Aakash Rana Magar, Ayush Tamrakar, Roshan Thapa

Student ID(s):

Student–Supervisor Meeting Log (Minimum 5 interactions are required):

Completed Tasks/ Achievement	Issues/ challenges	Next Week plan
1. Integrated Photon PUN2 networking: host creates room (Room_XXXX format), joiner enters room code to join (Networkmanager.cs, JoiningRoomManager.cs). 2. Implemented NetworkCar.cs with IPunObservable for position/rotation synchronization using lerp/slerp interpolation on remote cars. 3. Created NetworkCarManager.cs singleton (DontDestroyOnLoad) for network car instantiation via PhotonNetwork.Instantiate with automatic scene sync. 4. Built lobby system (LobbyManager.cs) with live player list, ready/unready toggle, and host-only start button that enables when all players are ready. 5. Implemented Google OAuth2 desktop flow (GoogleDesktopAuth.cs) with browser-based authentication, local TCP listener for redirect callback, token exchange, and user info fetch. 6. Added Photon custom properties for syncing player data: CarId, IsReady, FinishTime, TopSpeed, AverageSpeed, BestLap. 7. Implemented room custom properties for track selection and lap count synced across all players. 8. Created multiplayer join screen (JoiningRoomManager.cs) with room code input, error popups via AlertPopup, and status text feedback.	1. Position synchronization sends only position + rotation via IPunObservable causing jittery movement for remote cars in corners. 2. Remote cars have isKinematic=true disabling physics simulation, making car-to-car collisions unreliable. 3. Race countdown is local only; each client starts countdown at different times depending on scene load speed. 4. Lap/checkpoint progress not networked during race; remote player progress only communicated via FinishTime custom property after race finishes.	1. Develop in-race HUD displaying speed in KM/H and lap counter. 2. Build post-race results panel with multiplayer leaderboard sorted by finish time, best lap, top speed, average speed. 3. Implement pause menu with resume, restart, main menu, and live player list. 4. Add track selection UI with 3 track preview images and lap count dropdown.

Supervisor's Feedback:

Supervisor Name & Signature: Er. Saroj Giri _____

Date: _____